Health Care-Associated Infections (HAIs)

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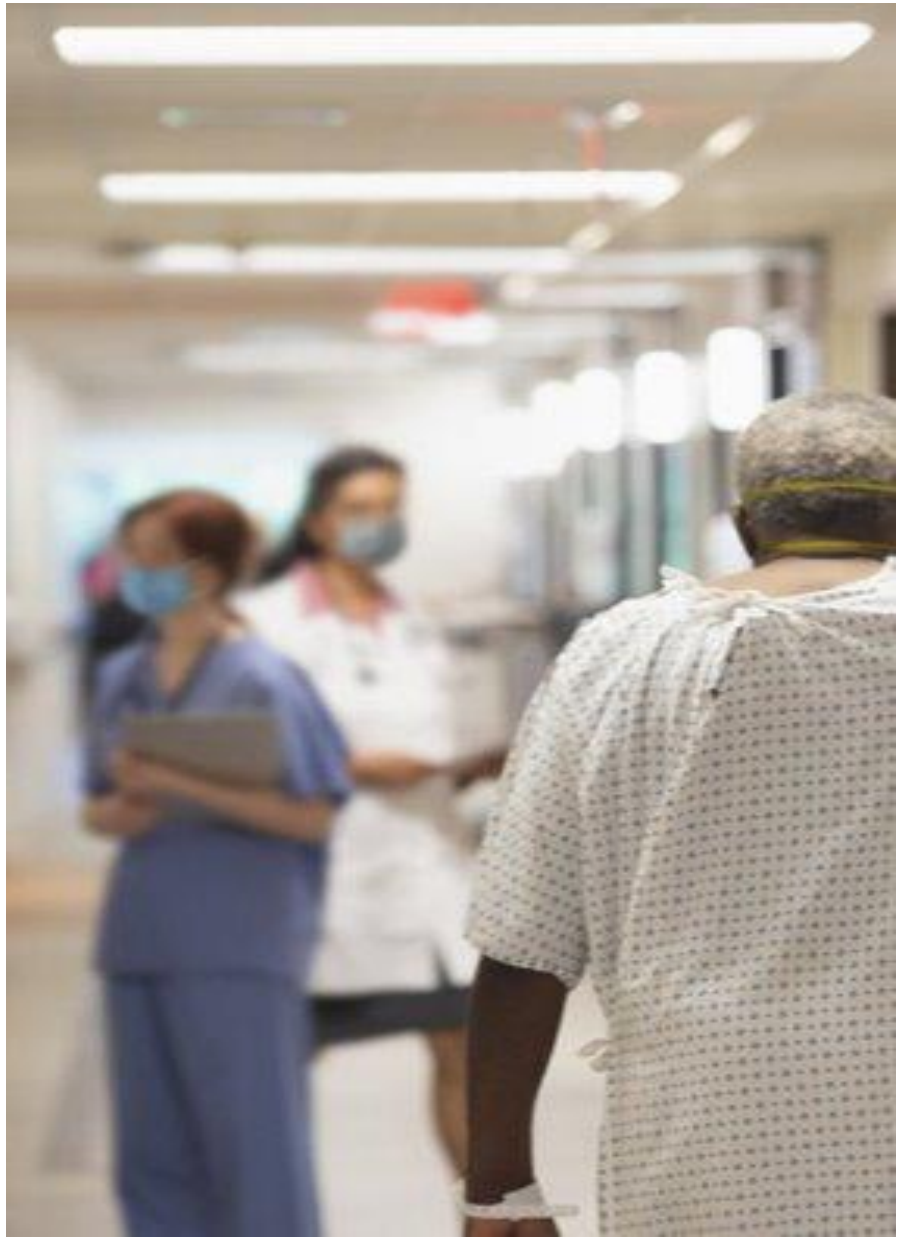
Learning Objectives of HAIs Study

- Define and identify the HAIs.
- Know the magnitude of the HAIs problem
- Describe the epidemiology of HAIs
- Identify the general and specific prevention and control measures for HAIs.



- HAIs = most frequent adverse event in healthcare delivery
- Affect hundreds of millions worldwide annually
- Prevalence:7–12% in developed countries
- 25–40% in developing countries

Definition

- Infections acquired in hospitals/health-care facilities
- Not present/incubating at admission
- May appear after discharge
- Includes occupational infections among staff

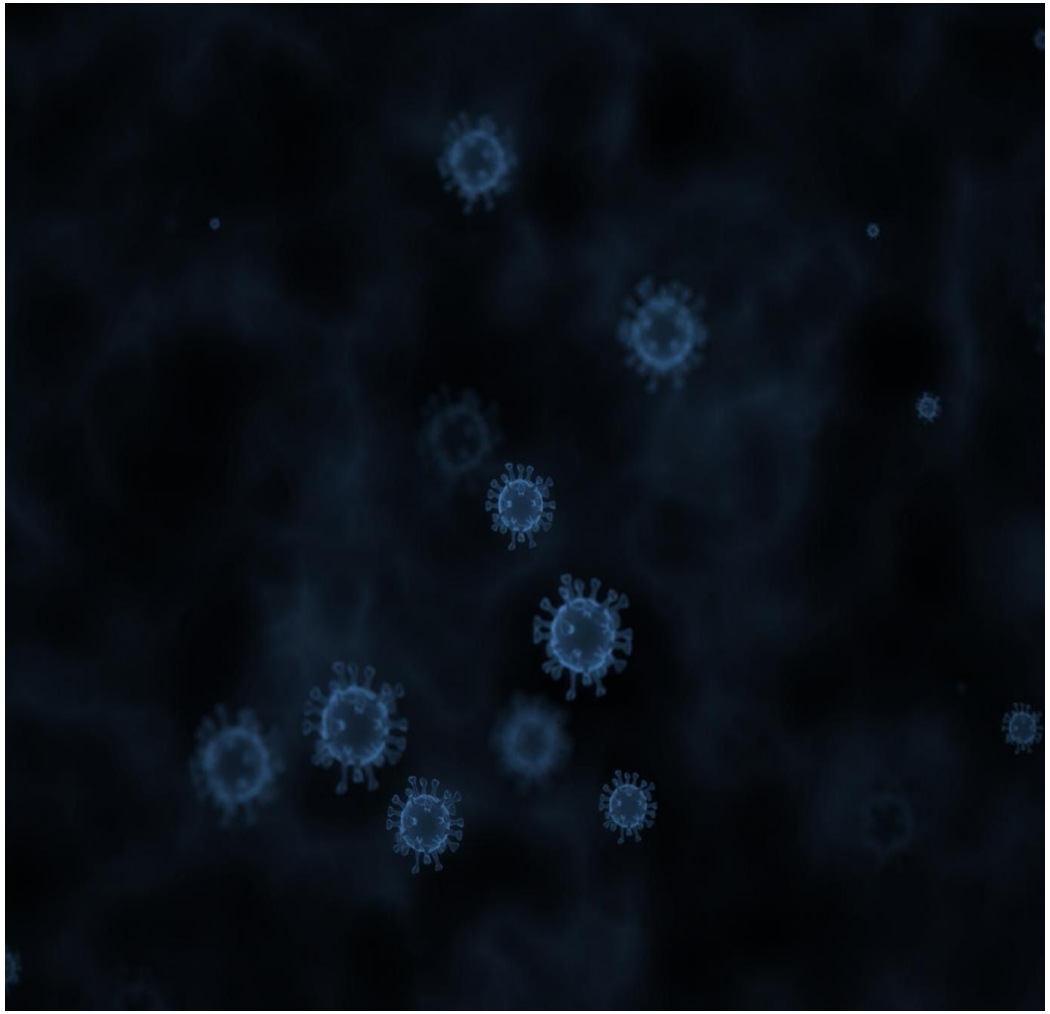
Causes

- **Failure to observe aseptic precautions during:**
  - Surgery
  - IV infusions
  - Catheterization
  - Wound dressings
  - Lumbar puncture, injections

Predisposing factors:

- Drug resistance in organisms
- Overcrowding of the hospitals
- Poor hospital environment
- Vulnerable groups: TB, diabetes, anemia, elderly, post-surgery, immunosuppressed

Etiological Agents Responsible for HAIs



- 50%: Staphylococci
- 45%: Gram-negative bacilli (E. coli, Klebsiella, Proteus, Pseudomonas, Salmonella, Shigella)
- 5%: Viruses, fungi, protozoa, rickettsiae
- HBV & HIV worsen situation

Major Categories and Clinical Types of HAIs

- **Urinary Tract Infections** which may be symptomatic and asymptomatic
- **Lower Respiratory Infections**
- **Postoperative Wound Infections** any surgical wound resulting in purulent discharge, is considered as HAI
- **Systemic Infections and Sepsis**
  - revealed by positive blood culture.

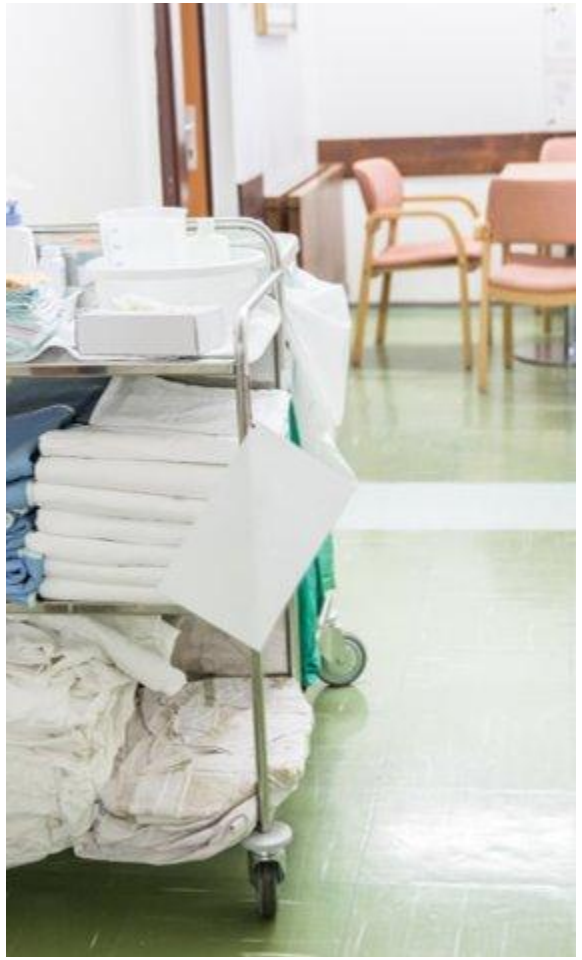
Sources and Reservoirs of Infection

- **Patient's own flora (autoinfection)**
- **Cross-Infection Among Patients** By direct contact, by air, by the staff via contaminated hands, clothing, or as carriers of microorganisms on mucosal surfaces.

Environmental source

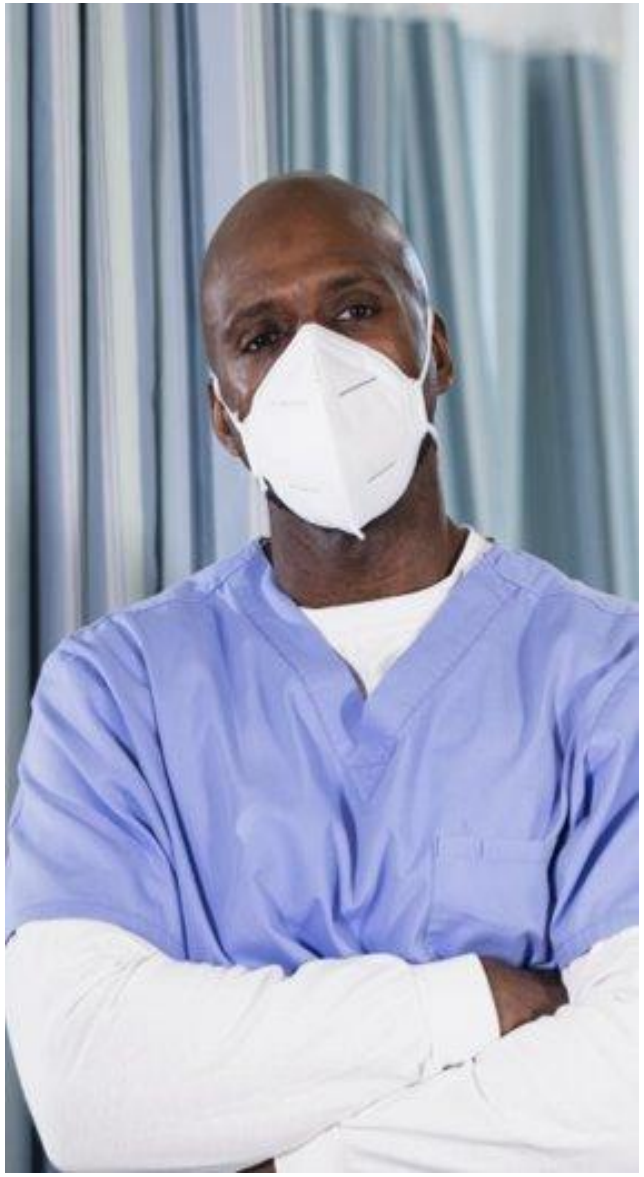
Environmental sources include contaminated **air**, **surfaces**, and **Contaminated Inanimate objects** .

Sources and Reservoirs of Infection



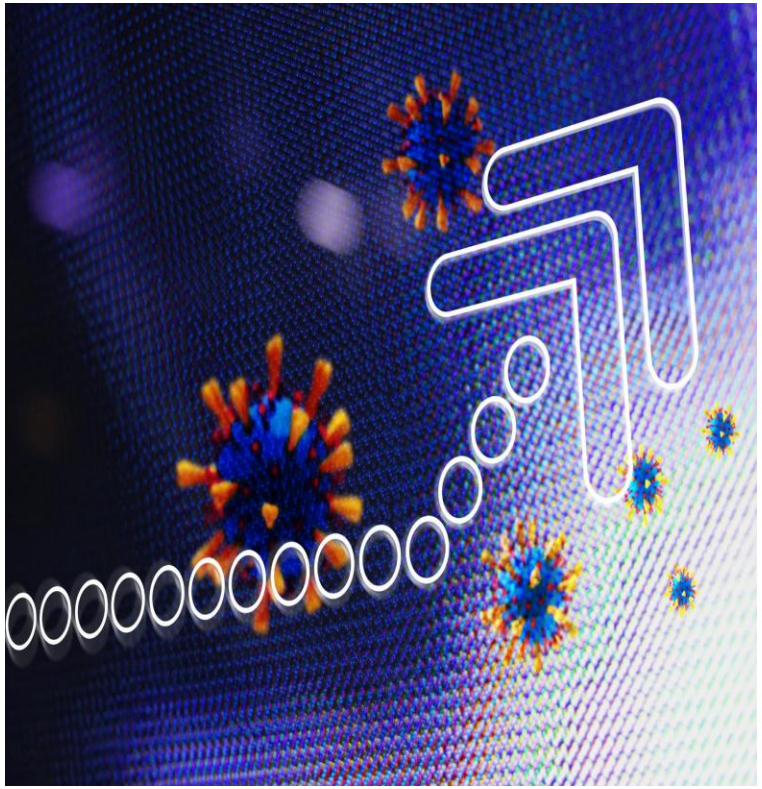
Inanimate objects may

- **Contaminated by the patients**: Hospital equipment, and medical equipment
- **Contaminated by the hands of any hospital staff** in any part of the hospital
- **Contaminated by visitors**
- **Contaminated by staff** who are ill or are carriers of microorganisms
- **Contaminated by food or contaminated water**
- **Contaminated by animals and insects.**



- **Thus, man occupies a central position:**
  - **a. As reservoir and source of microorganisms**
  - **b. As disseminator (communication routes)**
  - **c. As recipient or target, thus becoming a new reservoir.**
- **Age incidence: HAI can occur in any age group. But it is high in pediatric and geriatric age groups.**
- **Sex incidence: HAI is equal in both the sexes.**

Transmission Routes and Mechanisms



a. By contact: among the patients themselves by touch or with the doctor or staff.

b. By droplet infection

c. By infected dust: spread into the atmosphere while sweeping the wards or while bedmaking, etc.

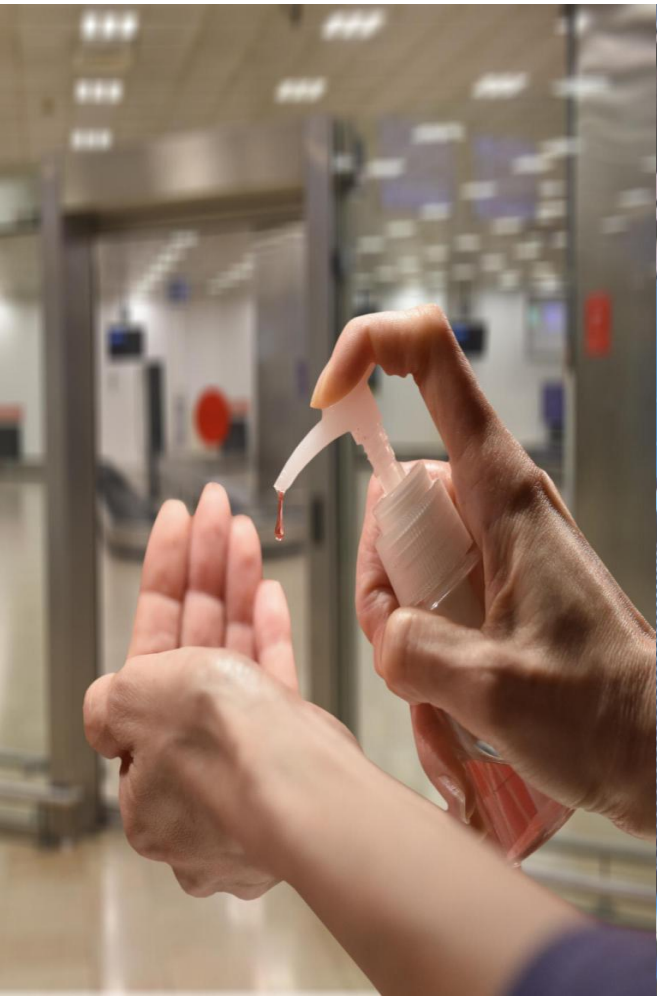
d.By fomites (inanimate objects)

e.By parenteral or percutaneous route during hospital procedures **(Iatrogenic).**

Prevention

- **Eliminate reservoirs/sources:**
  - **Care of the staff:** Staff with infections off duty
  - **Care of the patients:** Isolation of acute cases
  - **Treatment**: Correct antibiotic use
  - **Nursing care:**
    - Barrier nursing (gowns, gloves, masks)
    - Task nursing (specialized duties)

Breaking Transmission



1. Contact transmission by disinfecting hands with soap, water, or antiseptic lotion immediately after patient contact.

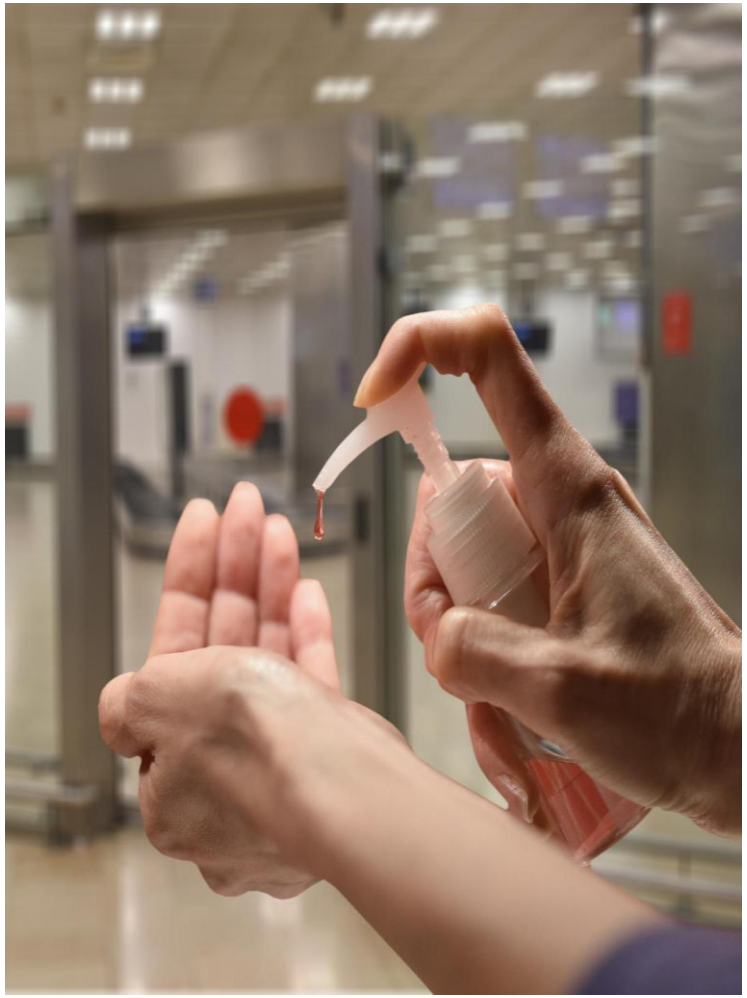
2. Isolation: Place infectious patients in isolation wards to limit spread.

3. Concurrent disinfection: Continuously disinfect patients' body fluids (urine, saliva, sputum, stool) and contaminated items (clothes, utensils, fomites).

4. droplet transmission:

- Avoid overcrowding
- Maintain 12-ft spacing between beds
- Ensure adequate lighting
- Provide good ventilation, especially cross-ventilation

Breaking Transmission



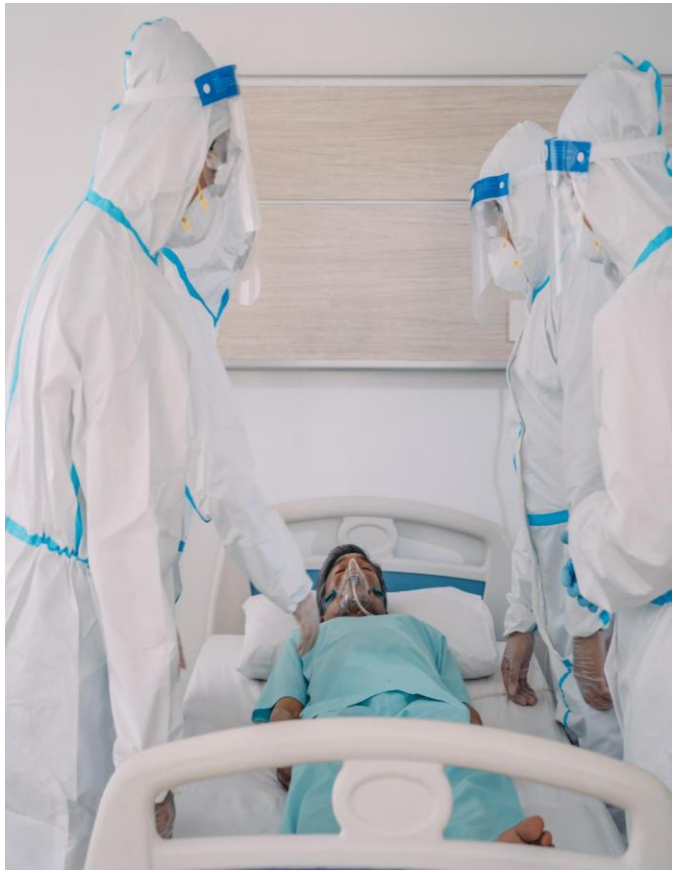
5. **Control of infected dust**: By vaccum cleaning of , by wet-mopping of the floor preferably with phenyl.

6. **Transmission by various hospital procedures** observing strict aseptic precautions.

7. **Hospital refuse** : best disposed off by incineration. Needles and syringes are disinfected before disposal.

8. **Hospital premises** should always be kept clean both inside and outside.

Protecting Susceptible Persons



- 1. Highly susceptible persons should be admitted preferably in the **isolation** wards.
- 2. They are **protected** with vaccinations, e.g. tetanus, hepatitis B, gas gangrene, etc.
- 3. **Universal blood and body fluid precautions** to be adopted by all the health care workers.

Administrative Measures

- Hospital Infection Control Committee
  - Surveillance of HAIs
  - Guidelines for high-risk procedures
  - Outbreak detection & control
  - Policies for admission, isolation, disinfection

Thank you